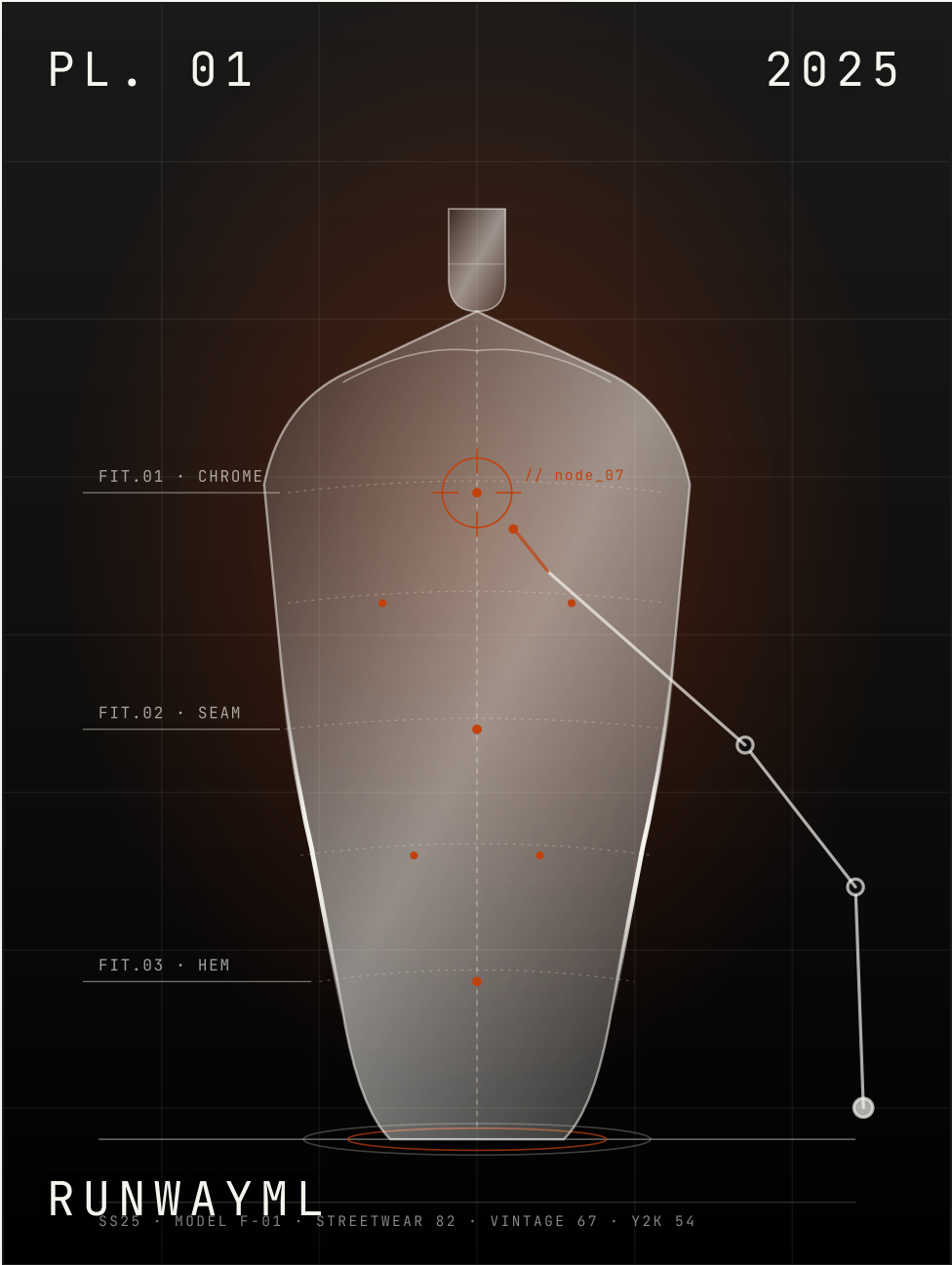


RunwayML

*An agentic AI for fashion trend forecasting,
end-to-end.*



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*Presented by
Tran Nguyen Ngoc Nhu*

02 Context.

A CATEGORY
MOVING IN REAL TIME.

Fashion trends move *faster* than the systems built to track them.

— a forecasting problem disguised as a styling one.

A PERSONAL STYLIST, ON CALL.

- “If I’m feeling sad today, what should I wear?”
- “What’s trending right now?”
- “What did I wear last week?”

Mood, memory, and the market — in one conversation.

- 01 Signals are scattered. Search interest, social mood, and personal wardrobes each tell a partial story.
- 02 Historical & real-time data rarely live together. Batch ETL lags; streaming alone has no memory.
- 03 Answers require synthesis. Stylists ask in plain language; the system must reason, retrieve, and predict.

03 Architecture.

STREAMING + BATCH,
UNIFIED FOR ONE AGENT.

SOURCES 01 INGESTION 02 WAREHOUSE 03 INTELLIGENCE 04 INTERFACE 05



04 The Agent.

ONE GRAPH,
THREE TOOLS, FULL REASONING.

A single LangGraph node decides *which signal to consult*, calls the right tool, and writes the answer.

CHATBOT.PY

```
class ChatbotState(TypedDict):
    messages: Annotated[list, add_messages]
    step_count: int
    approved: bool

def rag_node(state):
    augmented = llm.bind_tools(tools)
    response = augmented.invoke([sys, user])

    if response.tool_calls:
        # dispatch -> SQL / Pinecone
        results = dispatch(response.tool_calls)
        return augmented.invoke(
            [sys, user, response, *results])
    return response

workflow = StateGraph(ChatbotState)
workflow.add_node("rag", rag_node)
workflow.add_edge("rag", END)
chatbot = workflow.compile(
    checkpoint=MemorySaver())
```

TURN LOOP - STATEGRAPH



05 Conversation.

ONE QUESTION,
THREE TOOL CALLS, ONE ANSWER.

The agent *chooses its sources* mid-turn — historical, real-time, or semantic — then composes the reply.

- User asks a styling question
- LLM emits tool calls
- Snowflake & Pinecone respond
- LLM synthesizes final answer

USER "What's trending *right now*, and is it a real shift or just noise compared to the last twenty years?"

```
tool > query_realtime_data_snowflake()  
→ STREETWEAR • score 82 • today
```

```
tool > query_trend_data_snowflake()  
→ VINTAGE • highest avg popularity, 2004-2025
```

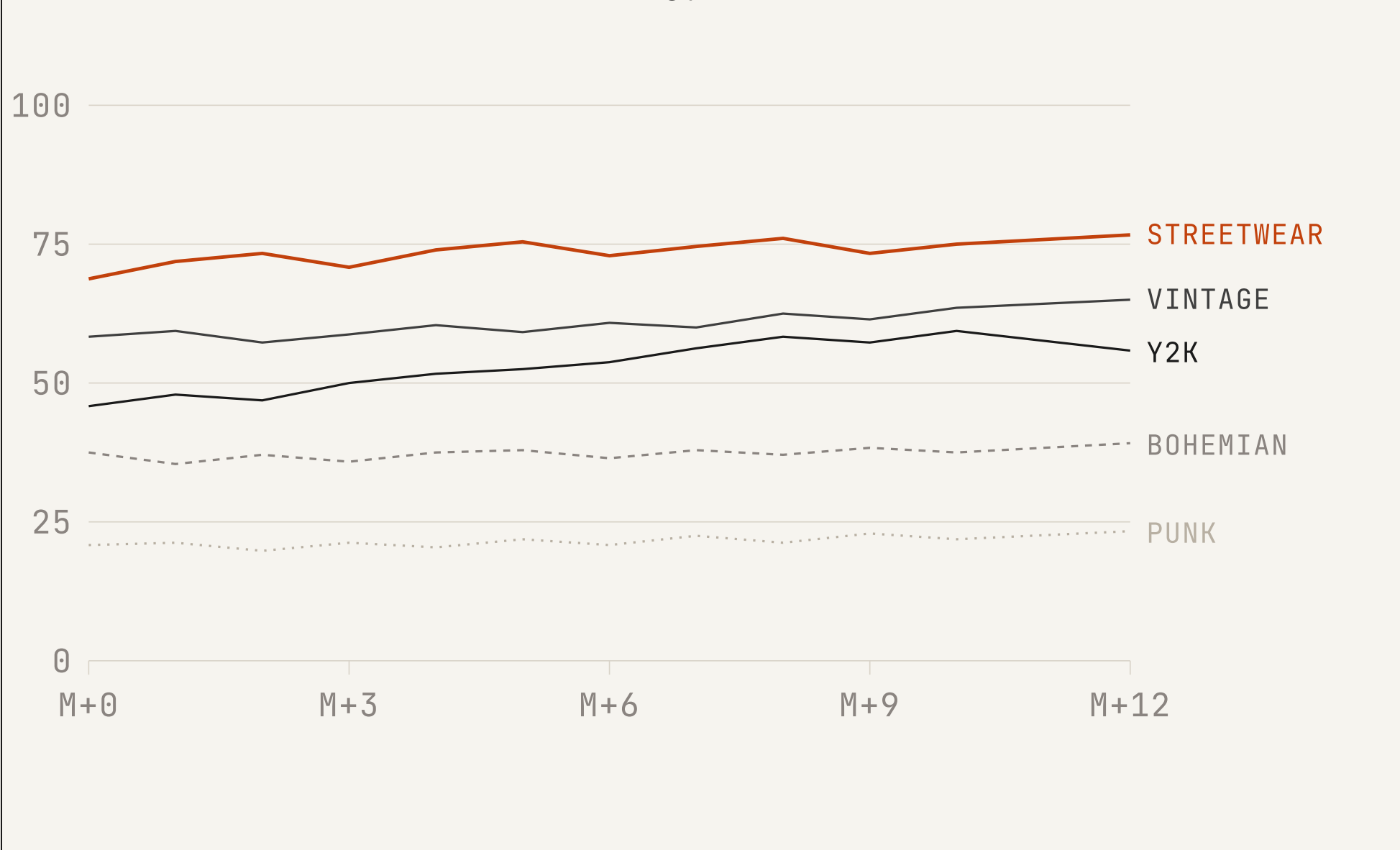
```
tool > search_similar_docs_pinecone("streetwear vs  
vintage")  
→ 2 chunks • editorial context retrieved
```

AGENT Today's leading signal is *streetwear* (82). Historically, *vintage* still tops the twenty-year average — so the spike is real, but reads as a seasonal swing, not a regime change.

06 Forecast.

A RANDOM FOREST READS
TWENTY YEARS AND PROJECTS TWELVE MONTHS.

FIG. 01 – PREDICTED POPULARITY, NEXT 12 MONTHS
Random Forest · 100 estimators · 1-month lag features



Streetwear stays at the top; vintage closes the gap by month twelve.

5 TREND CATEGORIES	20y TRAINING WINDOW
12mo FORECAST HORIZON	100 TREES PER MODEL

```
# lag features per category
for col in trend_cols:
    df[f"{col}_lag1"] = df[col].shift(1)

# one Random Forest per category
for trend in trend_cols:
    model = RandomForestRegressor(
        n_estimators=100, random_state=42)
    model.fit(X_train, y_train[trend])
    models[trend] = model

# autoregressive 12-step rollout
for _ in range(12):
    preds = [models[t].predict(curr)[0]
              for t in trend_cols]
    curr = np.array(preds).reshape(1, -1)
```

07 Colophon.

CREDITS, STACK,
AND WHERE TO FIND ME.

Built end-to-end — from
ingestion to *inference* to *interface*.

Designed & engineered by

Tran Nguyen Ngoc Nhu

DATA &
PIPELINES

Kafka · Prefect · Snowflake · dbt

MODELING

scikit-learn Random Forest · dimensional
modeling in dbt

AGENT

LangGraph · LangChain · Pinecone · GPT-
3.5

DELIVERY

Docker · Docker Compose · Rich CLI

REPOSITORY

github.com / notzdean / RunwayML